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Improving crop type mapping by integrating LSTM with temporal random masking and pixel-set spatial information

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Abstract

Accurate and timely crop type classification is essential for effective agricultural monitoring, cropland management, and yield estimation. Unfortunately, the complicated temporal patterns of different crops, combined with gaps and noise in satellite observations caused by clouds and rain, restrict crop classification accuracy, particularly during early seasons with limited temporal information. Although deep learning-based methods have exhibited great potential for improving crop type mapping, insufficient and noisy training data may lead them to overlook more generalizable features and derive inferior classification performance. To address these challenges, we developed a Mask Pixel-set SpatioTemporal Integration Network (Mask-PSTIN), which integrates a temporal random masking technique and a novel PSTIN model. Temporal random masking augments the training data by selectively removing certain temporal information to improve data variability, enforcing the model to learn more generalized features. The PSTIN, comprising a pixel-set aggregation encoder (PSAE) and long short-term memory (LSTM) module, effectively captures comprehensive spatiotemporal features from time-series satellite images. The effectiveness of Mask-PSTIN was evaluated across three regions with different landscapes and cropping systems. Results demonstrated that the addition of PSAE in PSTIN significantly improved crop classification accuracy compared to a basic LSTM, with average overall accuracy (OA) increasing from 80.9% to 83.9%, and the mean F1-Score (mF1) rising from 0.781 to 0.818. Incorporating temporal random masking in training led to further improvements, increasing average OA and mF1 to 87.4% and 0.865, respectively. The Mask-PSTIN significantly outperformed traditional machine learning and deep learning methods (i.e., RF, SVM, Transformer, and CNN-LSTM) in crop type mapping across all three regions. Furthermore, Mask-PSTIN enabled earlier and more accurate crop type identification before or during their developing stages compared to machine learning models. Feature importance analysis based on the gradient backpropagation algorithm revealed that Mask-PSTIN effectively leveraged multi-temporal features, exhibiting broader attention across various time steps and capturing critical crop phenological characteristics. These results suggest that Mask-PSTIN is a promising approach for improving both post-harvest and early-season crop type classification, with potential applications in agricultural management and monitoring.

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Abstract

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Keywords

Crop type mapping; Temporal random masking; Spatial information aggregation; Early season crop classification; Satellite image time series

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Introduction

The world's population has grown triple larger than in the mid-twentieth century (Lobell et al., 2013), rapidly increasing global food demands. Current agricultural practices to meet these rising food needs, such as expanding croplands, increasing fertilizer use, and intensifying cropping practices, may threaten the environment (Bennetzen et al., 2016). Sustainable agricultural management aims to balance food production and environmental protection (Tilman et al., 2011, van Dijk et al., 2021). Accurate and timely spatial distribution information on crop types is fundamental data for yield estimation, cropping patterns management, and pest control (Cai et al., 2018, van Loon et al., 2019), all of which are vital for sustainable agricultural practices (Asam et al., 2022, Xin et al., 2020). However, obtaining crop information through traditional field surveys and census data is time-consuming and expensive, especially for large-scale and historical monitoring (Becker-Reshef et al., 2010, Zhong et al., 2014).

The rapidly developing remote sensing techniques have been recognized as a cost-effective alternative for obtaining area and spatial distribution information of crop types (Hu et al., 2021, Lou et al., 2023, Skakun et al., 2017, Xia et al., 2022). Considerable efforts have been devoted to map crop types using satellite data and these approaches can be categorized into three groups: threshold-based methods, machine learning methods, and deep learning methods (Becker-Reshef et al., 2010, Boryan et al., 2011, Jiao et al., 2014, Weiss et al., 2020). Threshold-based methods identify crop types based on pre-determined classification rules and crop-specific phenological characteristics. For instance, Xiao et al. (2005) effectively classified paddy rice during the transplanting phase using the rule that the Land Surface Water Index (LSWI) is greater than the Enhanced Vegetation Index (EVI). However, the performances of these methods were highly dependent on the spatiotemporal stability of crop phenology. Due to the changes in climate and agricultural practices, the phenological characteristics of crop types may vary significantly across space and time, leading to poor generalizability and transferability of pre-determined classification rules across different regions and years. Machine learning methods, such as random forest, decision tree, and support vector machine, have been successfully applied in classifying different crop types over large areas due to their ability to automatically learn relationships between satellite-derived features and crop types (Blickensdörfer et al., 2022, Fowler et al., 2020, Hu et al., 2021, Yoon and Kim, 2020). Nevertheless, these methods typically rely on shallow and hand-crafted features and struggle to capture complex temporal patterns during crop growth, limiting their performance and generalizability for mapping crop types (Zhong et al., 2019).

In contrast, deep learning methods have proven effective in extracting deep representative features from multi-temporal remote sensing data for crop type mapping (Asadi and Shamsoddini, 2024, Cai et al., 2023b, Cai et al., 2024, Tang et al., 2024, Wang et al., 2023). Among these methods, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are widely used approaches according to previous studies (Wang et al., 2022, Zhang et al., 2024). Specifically, CNNs primarily extract texture information from images in the spatial domain (Zhong et al., 2019), while RNNs are specialized in modeling temporal dependencies within sequential data, making them particularly effective in capturing the phenological dynamics of crops from time-series satellite data (Feng et al., 2023, Xu et al., 2023). Long short-term memory (LSTM) units, a variant of RNN, have shown a superior ability to memorize long-term temporal information by introducing a gated mechanism, enabling the successful applications of LSTM-based methods in large-scale crop type mapping (Wang et al., 2020,

Zhong et al., 2019). Crisóstomo De Castro Filho et al. (2020) compared the performance of LSTM and bidirectional LSTM (BiLSTM) in rice mapping in Brazil, finding that BiLSTM exhibited better results by learning more comprehensive information from both forward and backward directions. Xu et al. (2020) further improved LSTM's ability to explore the temporal features by incorporating the attention mechanism, significantly outperforming traditional machine learning models in mapping corn and soybean distribution over large areas. Despite these advancements, the practical application of time-series satellite data is often influenced by noise and gaps due to cloud cover and rain, leaving it challenging to explore robust and invariant crop features (Zhao et al., 2021). The challenge is exacerbated by the difficulty in collecting a substantial number of samples, which can cause the model to overfit a specific temporal pattern or noise present in the training data, ultimately reducing classification performance (Salman and Liu, 2019, Zhao, 2017). Moreover, early-season crop mapping, which refers to the classification of crop types using limited satellite observations during the early stages of the growing season, generally shares similar classification methods with post-harvest crop mapping but uses fewer available satellite images (Wang et al., 2024, You and Dong, 2020). Nevertheless, it is particularly challenging compared to post-season classification due to the less temporal information available for analysis (McNairn et al., 2014, Potgieter et al., 2010, Zhang et al., 2021).

To mitigate these challenges, several studies have explored reconstructing cloud-free time-series observations through temporal interpolation and multi-source data integration, aiming to capture critical phenological information about crops and improve classification accuracy (Wang et al., 2024, Zhou et al., 2024). However, recent studies suggest that these interpolation processes may not reliably reconstruct time series, offering limited improvement in crop type mapping and increasing computational costs (Che et al., 2024, Zhao et al., 2021). By contrast, data augmentation techniques have been demonstrated to increase the training data variability and improve models' robustness against noise and gaps. For instance, randomly removing contiguous parts of input images during data augmentation can significantly improve models' performance in handling occlusion issues of image classification (DeVries and Taylor, 2017, Zhong et al., 2020). The combination of different augmentations to process audio sequence data, such as frequency masking and time masking, has been proven to improve speech recognition performance even with limited training data (Alex et al., 2023, Park et al., 2019). As for time-series satellite image analysis, Chen et al. (2022) developed a novel imputation-BiLSTM model that introduced a missing data imputation task, jointly trained with crop classification, to address incomplete time-series satellite data caused by cloud contamination. Garnot et al. (2022) adopted a temporal dropout strategy that randomly removed observations from multi-modal time-series data during training stages to prevent the model from overly relying on a single modality. Inspired by this, Pham et al., 2024a, Pham et al., 2024b introduced random observation selection to simulate the variations and sparsity of temporal data, thereby improving the models' transferability. Although these augmentation approaches have made significant progress, they were mainly tested in single agricultural landscapes. The generalizability of these methods across diverse crop types and landscapes remains unexplored. Therefore, it is critical to develop an effective data augmentation strategy to address overfitting in limited and low-quality datasets and improve the model's ability to capture crop phenology dynamics.

Furthermore, traditional classification methods were mainly implemented at the pixel level, which not only were easily affected by the "salt-and-pepper" problem but also neglected spatial features during crop growth caused by varying agricultural management practices and planting patterns (e.g., furrows or crop rows) (Jiao et al., 2014, Lin et al., 2022, Valero et al., 2021, Wei et al., 2023). Although the CNNs have been widely used for extracting spatial features from satellite images, their heavy reliance on texture may limit their effectiveness in leveraging spatial information from medium-resolution (10–30 m) satellite images, which may be too coarse to depict the agricultural textural details (Geirhos et al., 2018, Garnot et al., 2019). Recent studies have pointed out that viewing medium-resolution images as unordered sets of pixels and using pixel-set encoders to extract statistical features of spectral distribution within agricultural field parcels can be more effective than traditional CNNs in crop type mapping

(Garnot et al., 2020, Weilandt et al., 2023, Yuan et al., 2023). These methods typically depend on agricultural field parcel data as a spatial constraint to ensure that pixels in the pixel-set share the same category information. Despite the increasing availability of open-access agricultural field parcel datasets and the rapid development of field delineation methods (Cai et al., 2023a, Meyer et al., 2020), obtaining timely and accurate field parcel data remains challenging, particularly in smallholder farming regions like southern China. However, applying pixel-set methods directly to image patches can lead to inferior results due to the potential inclusion of features from different crops within the patch. Therefore, effectively integrating information from spatially neighboring pixels is crucial for capturing spatial information of crops and reducing noise effects in individual pixels, especially in areas characterized by small and heterogeneous landscapes.

In this study, we propose a deep learning method named Mask Pixel-set SpatioTemporal Integration Network (Mask-PSTIN), which leverages the spectral-spatial–temporal information of the time-series satellite images to improve the accuracy of crop type classification. The major contents of our study can be summarized in three aspects: (1) a temporal random masking augmentation strategy, which selectively masks observations at certain time steps from the time-series data, encouraging the model to learn more generalizable and robust features to avoid overfitting; (2) a novel pixel-set aggregation encoder that adaptively aggregates the spectral information from spatially neighbored pixels, integrated with the LSTM units and a temporal attention mechanism to explore the comprehensive spectral-spatial–temporal information from multi-temporal images, improving the accuracy of post-harvest and early-season crop type mapping; (3) based on gradient backpropagation methods, the feature importances were analyzed to interpret the advantages of Mask-PSTIN. Furthermore, we evaluated the effectiveness and generalization of Mask-PSTIN with experiments across three regions characterized by diverse agricultural landscapes and cropping patterns.

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The overview of the Mask-PSTIN model

The framework of the developed Mask-PSTIN is presented in Fig. 1. Mask-PSTIN consists of a novel classification model (PSTIN) and a data augmentation technique involving temporal random masking. Specifically, PSTIN combines a pixel-set aggregation encoder (PSAE) with a BiLSTM and a temporal attention (AtBiLSTM) module to capture the spatiotemporal patterns of time-series satellite image patches. The random masking technique augments the limited training data by selectively masking observations …

Study area

To evaluate the performance of our developed Mask-PSTIN model in post-harvest and early-season crop mapping, we selected three heterogeneous regions with different cropping patterns and agricultural landscapes (Fig. 5). The phenological calendars of the main crop types across the three study regions are shown in Fig. 6.

Region A (44.6°N – 46.8°N, 2°E – 4.5°E) is Auvergne situated in central France, including four departments: Allier, Puy-de-Dôme, Cantal, and Haute-Loire. It has a temperate …

Ablation experiment of Mask-PSTIN in crop classification

Table 2 presents the OA and mF1 of four models (i.e., LSTM, Mask-LSTM, PSTIN, and Mask-PSTIN) in crop type classification across the three study regions. The Mask-PSTIN model demonstrates remarkable robustness and superior performance, achieving the highest average OA of 87.4% and mF1 of 0.865. The LSTM model, serving as the baseline, derived the lowest average OA of 80.9% and mF1 of 0.781. Compared to basic LSTM, the Mask-LSTM model that trained with temporal random masking augmentation …

Explanation of the Mask-PSTIN advantages in crop type mapping

To further understand the efficacy of Mask-PSTIN in classifying different crops, we introduced gradient propagation to assess the importance of input time-series features in deep learning models (Tang et al., 2024, Xu et al., 2021). Since the deep neural networks are fully differentiable functions designed to model the correlations between input time-series features $[X_1, X_2, \ldots X_t]$ and the target crop type y, the gradient can be backpropagated to each element X_i within the input feature …

Conclusion

This study developed the Mask-PSTIN that comprises temporal random masking technique and a novel PSTIN model for mapping both post-harvest and early-season crop types. Specifically, the temporal random masking technique improves variability of limited training data by selectively masking observations at certain time steps in the multi-temporal satellite images during the training, enforcing the model to learn more robust and invariant features of crop types. PSTIN combines a novel pixel-set …

CRediT authorship contribution statement

Xinyu Zhang: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Zhiwen Cai:** Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Qiong Hu:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jingya Yang:** Validation, Methodology. **Haodong Wei:** Validation, Data curation. **Liangzhi You:** Writing – review & editing, Supervision, Conceptualization. **Baodong Xu:** Writing – review & editing, Supervision, Methodology, Funding …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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